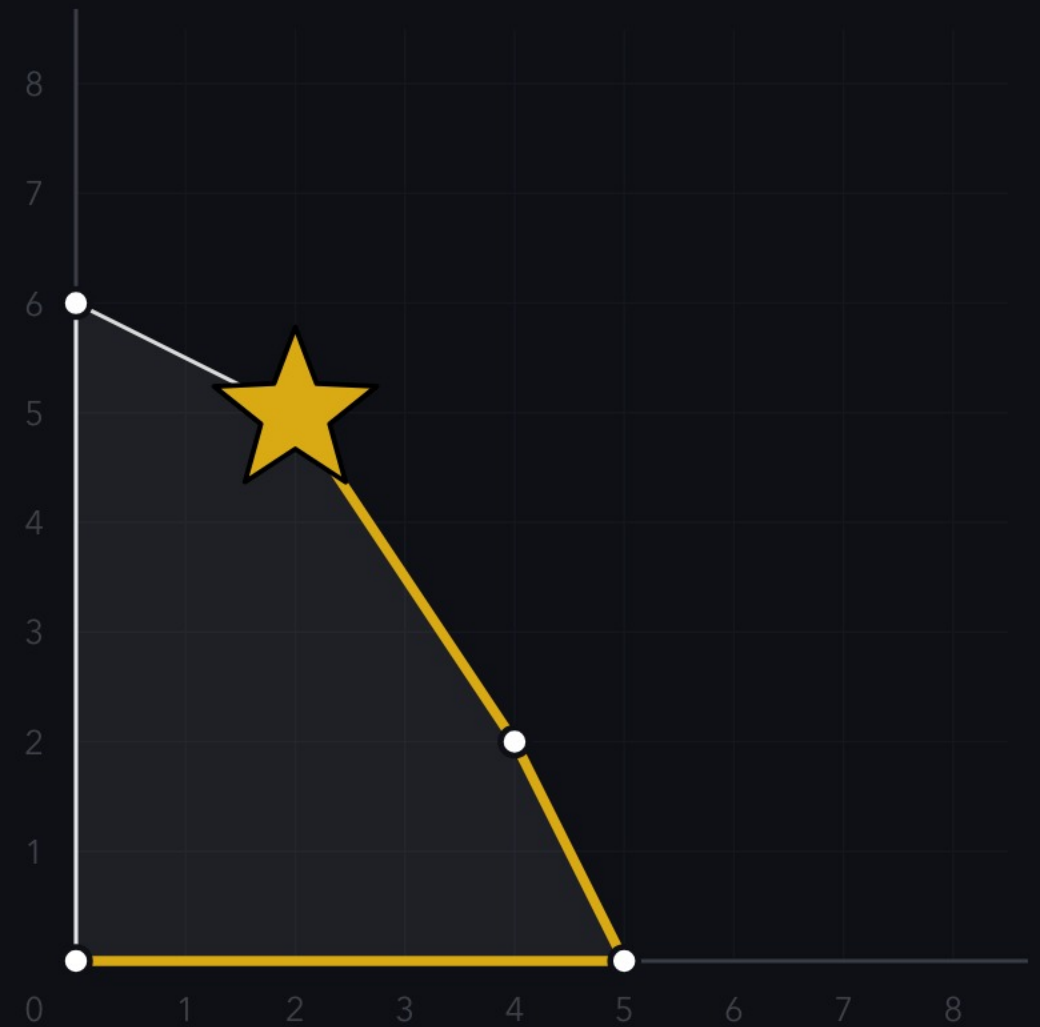


Walking the Corners

The simplex method, duality, and the art of choosing a pivot

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Made for: Optimal Lab · Johns Hopkins University



Linear programs (LPs)

Making the most of limited resources



Business decisions

decide how much of each product
 to make

limited by materials, machine time, staff
 hours

goal maximize profit

Our running example: a small bakery.



Radiation therapy planning

decide the intensity of each beam

must meet the prescribed dose to the
 tumor

goal as little dose to healthy
 organs as possible

We assume dose at each point is linear in the
beam intensities.



Scheduling appointments

decide which patients go in which
 slots

limited by clinicians, rooms, hours in
 the day

goal minimize patient wait time

Whole-number decisions: integer programs,
which solve many LPs along the way.

Same shape every time: $\underbrace{\max c^T x}_{\text{goal}} \quad \text{s.t.} \quad \underbrace{Ax \leq b}_{\text{limited resources}}, \quad \underbrace{x \geq 0}_{\text{decisions}}$

A small business example: our bakery

Two products, three limited resources

	bread	cake	available
■ flour	2	1	10
■ oven hours	3	2	16
■ labor hours	1	2	12
profit / batch	9	8	

\$ per batch. Units are made up; I'm not actually a baker.

x_1 = bread batches, x_2 = cake batches decisions

$$\max z = 9x_1 + 8x_2 \quad \text{profit}^*$$

$$\text{s.t. } 2x_1 + x_2 \leq 10 \quad \text{flour}$$

$$3x_1 + 2x_2 \leq 16 \quad \text{oven hours}$$

$$x_1 + 2x_2 \leq 12 \quad \text{labor hours}$$

$$x_1, x_2 \geq 0 \quad \text{no negative items (why?)}$$

$\implies$ would imply we can *unmake* less profitable items and harvest their flour, labor and oven hours for more profitable products

*recall: profit = revenue – cost

Our bakery in the general form

The formula from slide 2, filled in with the bakery's numbers

	bread	cake	available
■ flour	2	1	10
■ oven hours	3	2	16
■ labor hours	1	2	12
profit / batch	9	8	

$$\max c^T x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0$$

profit per batch

$$c^T = (9 \quad 8)$$

resources used per batch
rows: resources · columns: products

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 2 \end{pmatrix}$$

resources available

$$b^T = (10 \quad 16 \quad 12)$$

$$\max (9 \quad 8) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \text{s.t.} \quad \begin{pmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 10 \\ 16 \\ 12 \end{pmatrix}, \quad x \geq 0$$

Every constraint cuts the plane in half

The plans that satisfy all of them form the *feasible region*

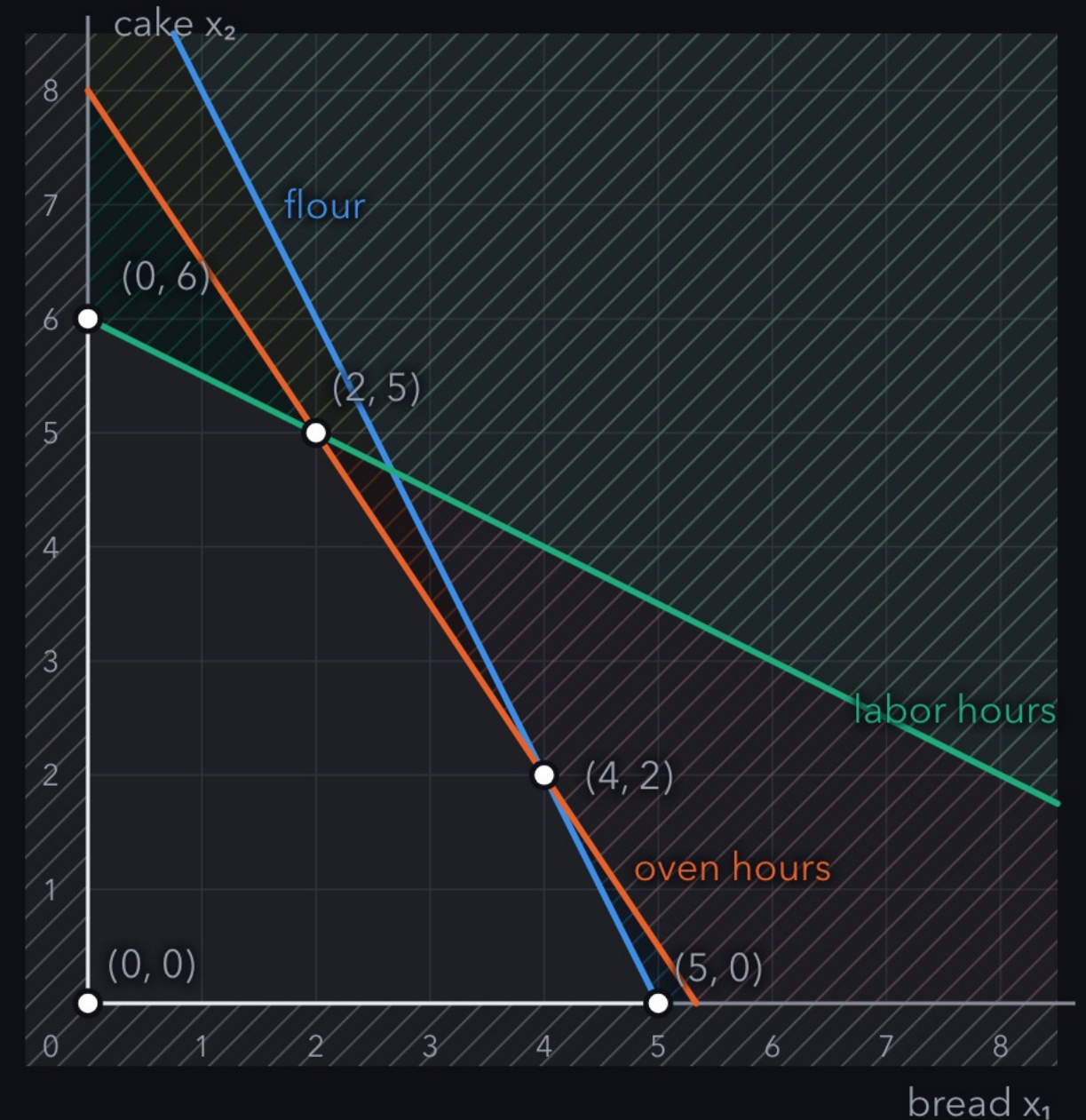
$x_1, x_2 \geq 0$: the positive quadrant

$2x_1 + x_2 \leq 10$ x_1 -intercept (5, 0), x_2 -intercept (0, 10)

$3x_1 + 2x_2 \leq 16$ x_1 -intercept ($\frac{16}{3}$, 0), x_2 -intercept (0, 8)

$x_1 + 2x_2 \leq 12$ x_1 -intercept (12, 0), x_2 -intercept (0, 6)

Five **corners**. Each one is where two constraints meet.



Slide the profit line

All plans with the same profit lie on one line

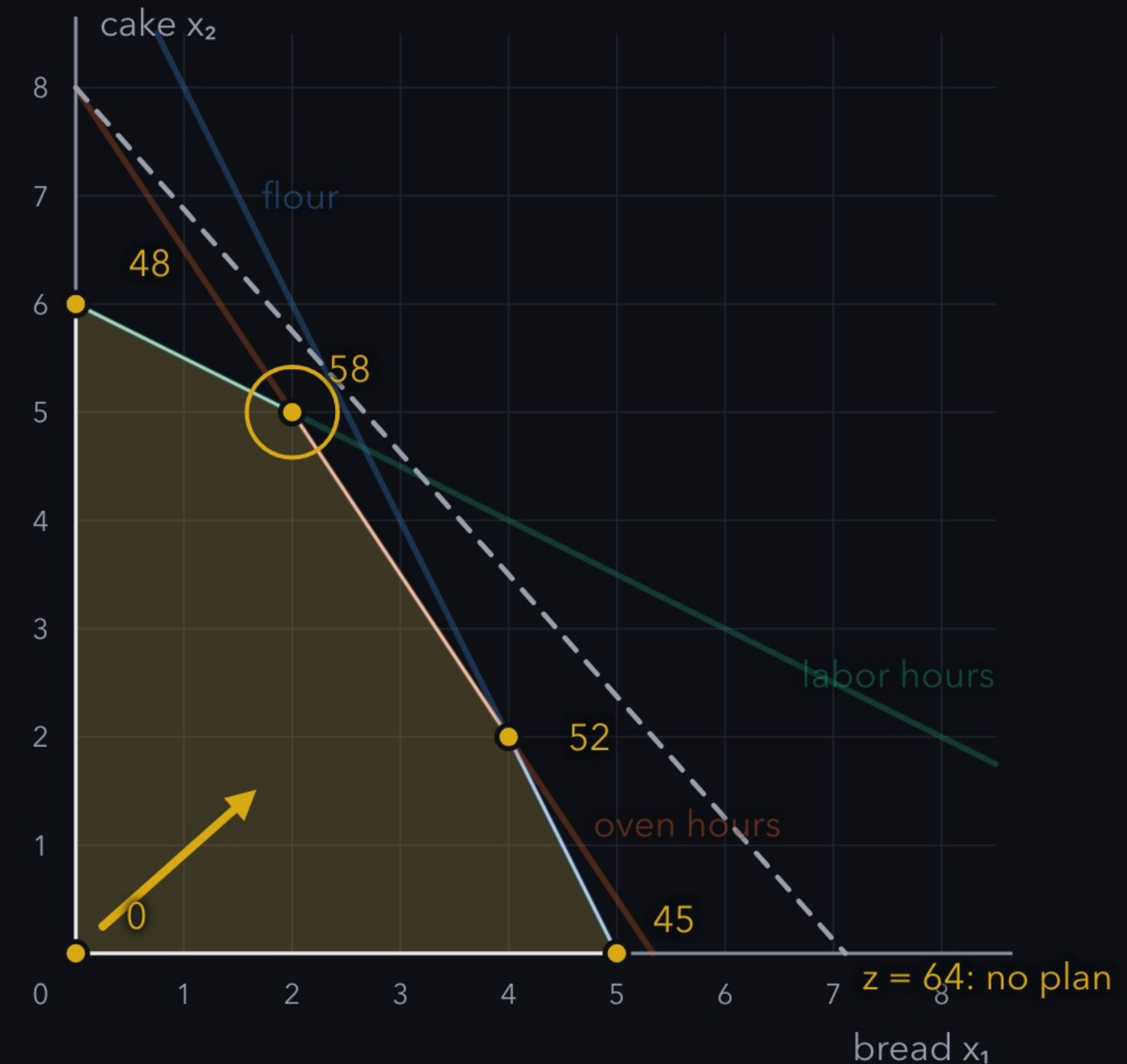
$$9x_1 + 8x_2 = z$$

$$z = 64$$

Push the line in the direction of the **arrow**, as far as the region allows.

The last plan it touches is a **corner**: (2, 5), profit **58**.

Any further and no plan is left.



Always a corner... unless the profit line is parallel to an edge

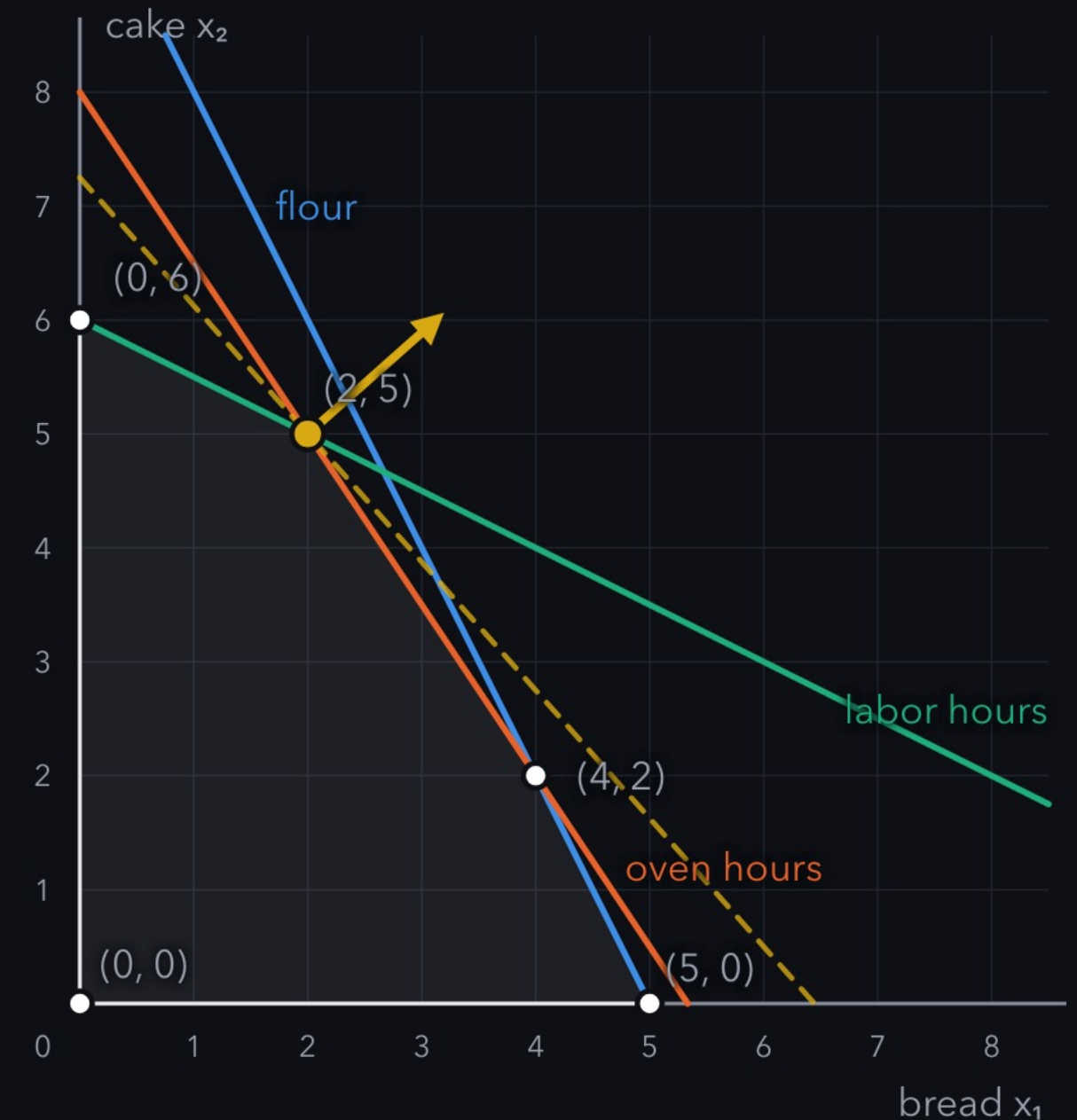
Turn the profit direction and watch the best corner

$$\max z = 9.0x_1 + 8x_2$$

Best plan: one corner, **(2, 5)**, earns $z = \$58.0$.

bread profit 9.0

Parallel to an edge $\Rightarrow$ the **whole edge** ties. Its two ends are still corners, so **checking corners is always enough**.



Optimality: how do we prove 58 is the best?

Plan *feasibility* is easy to check

Bake $x = (2, 5)$:

$$2 \cdot 2 + 5 = 9 \leq 10 \checkmark$$

$$3 \cdot 2 + 2 \cdot 5 = 16 \leq 16 \checkmark$$

$$2 + 2 \cdot 5 = 12 \leq 12 \checkmark$$

$$\text{Profit: } 9 \cdot 2 + 8 \cdot 5 = 58.$$

"Nothing beats it" is harder

Check every corner? Fine here: there are 5.*

A business with 50 products and 50 resources: up to $\binom{100}{50} \approx 10^{29}$ corners.

We want a short proof that anyone can check.

* $\binom{5}{2} = 10$ possible corners in our 2D problem, but only 5 are feasible

LPs have a "built in" proof, in the form of an upper bound. Given n variables and $m \geq n$ constraints, the best plan sits where n of the constraints meet, and the proof is built from those same constraints. When a valid upper bound meets a valid lower bound, we are at optimality, and the principle of **strong duality** guarantees that, whenever there is a best plan, they meet.

Duality and bounds*

$$\max 9x_1 + 8x_2$$

- A feasible solution to our original model (called the *primal* model) is a **lower bound**.

$$\begin{array}{ccccccc} 9(0) + 8(0) = 0 & \rightarrow & 9(5) + 8(0) = 45 & \rightarrow & 9(4) + 8(2) = 52 & \rightarrow & 9(2) + 8(5) = 58 \\ \text{lower bound} & & \text{new lower bound} & & \text{new lower bound} & & \text{new lower bound} \end{array}$$

- Any function with coefficients $\geq$ our objective function's coefficients is $\geq$ our objective function, for any value of $x \geq 0$.

e.g., $9x_1 + 8x_2 \leq 10x_1 + 8x_2$

- We can thus use constraints, that are all **true** for this problem, to bound our objective function **from above**:

$$2x_1 + x_2 \leq 10 \quad (\times 8) \quad \Rightarrow \quad 16x_1 + 8x_2 \leq 80$$

$$z = 9x_1 + 8x_2 \leq 16x_1 + 8x_2 \leq 80$$



* Adapted from A. Majumdar, ["Duality: the unsung hero of optimization explained through an example"](#), Towards Data Science (2021).

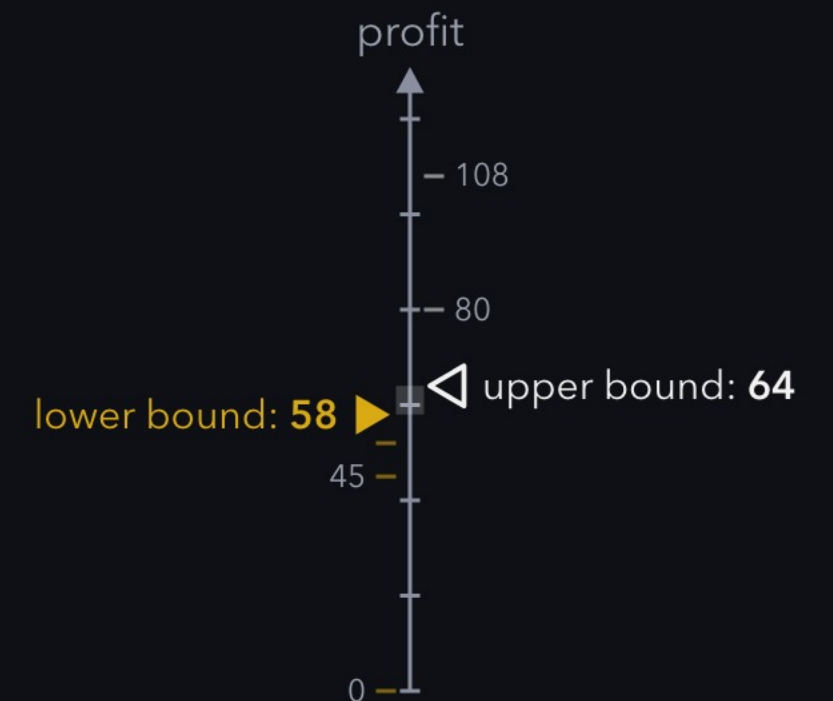
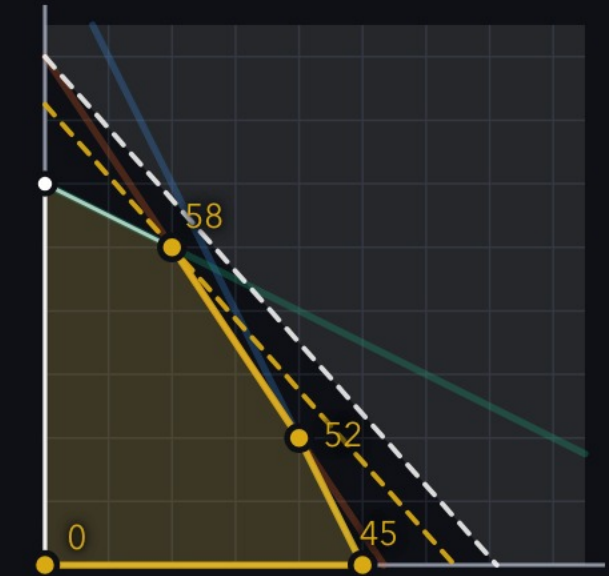
Understanding dual variables

Given our primal model, we can assign a *dual variable* y to each constraint

$$\begin{aligned} \max \quad & 9x_1 + 8x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 10 \quad (y_1) \\ & 3x_1 + 2x_2 \leq 16 \quad (y_2) \\ & x_1 + 2x_2 \leq 12 \quad (y_3) \\ & x_1, x_2 \geq 0 \end{aligned}$$

- Our first dual guess from the previous slide was $y^\top = (8, 0, 0)$, with $z \leq 80$.
- Constraint 2 would give us $y^\top = (0, 4, 0)$: $z \leq 64$; constraint 3, $y^\top = (0, 0, 9)$: $z \leq 108$. So **64 is our best upper bound** so far.
- We can also take combinations of the constraints, but they need to remain $\geq$ our coefficient values.

Dual variables $y \geq 0$ whose mix covers the profit form a **valid dual**, with bound $10y_1 + 16y_2 + 12y_3$.



The bound game: add up the constraints

Pick a dual variable ≥ 0 for each resource. If the mix covers the profit, it is a valid dual: it caps the profit.

0 × (2x₁ + x₂ ≤ 10)

2.5 × (3x₁ + 2x₂ ≤ 16)

1.5 × (x₁ + 2x₂ ≤ 12)

flour

oven

labor

sum: $9x_1 + 8x_2 \leq 58$

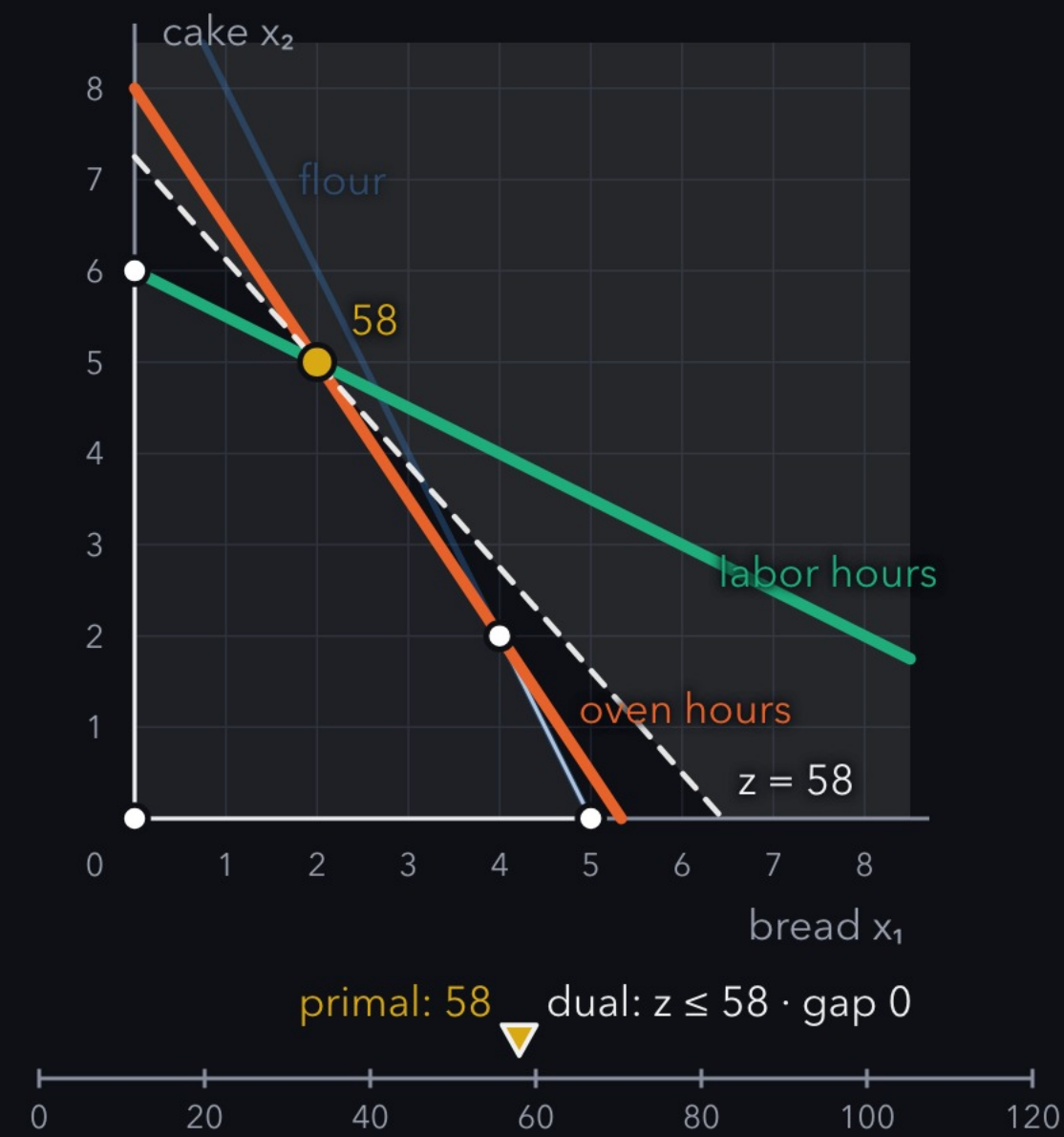
	sum covers	profit	dual slack (covers – profit)
bread x ₁	9	9	0 ✓
cake x ₂	8	8	0 ✓

✓ **Valid dual: z ≤ 58.** The primal (2, 5) earns exactly 58, so it is **optimal**.

reset

2.5 oven + 1.5 labor

Cover **both exactly**: $3y_2 + y_3 = 9, 2y_2 + 2y_3 = 8 \Rightarrow y_2 = 2.5, y_3 = 1.5$. Flour gets 0: it has a spare unit, so weight on flour would raise the bound but not the profit.



Formalizing the dual

The bound game as an LP: choose $y \geq 0$ so the mix covers every product's profit, with the smallest bound

PRIMAL

$$\begin{aligned} \max \quad & 9x_1 + 8x_2 \\ \text{s.t.} \quad & \boxed{\begin{aligned} 2x_1 + x_2 &\leq 10 & (y_1) \\ 3x_1 + 2x_2 &\leq 16 & (y_2) \\ x_1 + 2x_2 &\leq 12 & (y_3) \end{aligned}} \\ & x_1, x_2 \geq 0 \end{aligned}$$

$$A^T = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

2×3

DUAL

$$\begin{aligned} \min \quad & 10y_1 + 16y_2 + 12y_3 \\ \text{s.t.} \quad & \boxed{\begin{aligned} 2y_1 + 3y_2 + y_3 &\geq 9 \\ y_1 + 2y_2 + 2y_3 &\geq 8 \end{aligned}} \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

min: we want the **smallest** upper bound

- In other words, the sum of all the coefficients in the first column times their dual variables must be $\geq$ the objective function coefficient.
- And for every unit of dual value, one unit of the b vector is added.
- The primal's columns became the dual's rows: the dual uses A^T .

Primal and dual, in general

PRIMAL

$$\max \mathbf{c}^\top \mathbf{x}$$

$$\text{s.t. } \mathbf{A}\mathbf{x} \leq \mathbf{b} \quad (y)$$

$$\mathbf{x} \geq 0$$

n variables, m constraints

$$\mathbf{A}^\top = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{mn} \end{pmatrix}$$

$n \times m$

DUAL

$$\min \mathbf{b}^\top \mathbf{y}$$

$$\text{s.t. } \mathbf{A}^\top \mathbf{y} \geq \mathbf{c}$$

$$\mathbf{y} \geq 0$$

m variables, n constraints

- Transposing swaps rows and columns: $\mathbf{A}$ is $m \times n$, $\mathbf{A}^\top$ is $n \times m$.
- Each column j , times the dual variables, must cover its objective coefficient: $\sum_i a_{ij} y_i \geq c_j$ for every j , that is, $\mathbf{A}^\top \mathbf{y} \geq \mathbf{c}$.
- Every unit of y_i adds b_i to the bound: $\sum_i b_i y_i = \mathbf{b}^\top \mathbf{y}$, and we want the smallest bound.
- LPs have **weak duality**: for every feasible $\mathbf{x}$ and $\mathbf{y}$, $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$. They also have **strong duality**: if the primal has an optimal solution, so does the dual, and at optimality $\mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y}$.

Common names and interpretations for parts of the dual

Dual variable = shadow price

y_1, y_2, y_3 : what one more unit of each resource is worth. At the optimum, $y = (0, 2.5, 1.5)$.

Best profit = best bound = $10y_1 + 16y_2 + 12y_3$, so one more oven hour adds y_2 , as long as the same dual stays best: 17 hours $\Rightarrow 58 \rightarrow 60.5$, that's $+2.5$. One more unit of flour $\Rightarrow +0$ (we already have spare).

Dual slack = reduced cost

Each product has a dual constraint: the value of the resources it uses must cover its profit (bread: $2y_1 + 3y_2 + y_3 \geq 9$). Its slack is the product's **reduced cost**: value of the resources it uses – its profit.

Negative $\Rightarrow$ the prices undervalue it $\Rightarrow$ worth making more. At $(0, 0)$ every price is 0, so $d_1 = -9$ and $d_2 = -8$: both are worth making.

Two readings of the same equations

The primal and the dual are **two readings of the same equations**: at the best corner, the tight constraints form one square system.

primal
by rows $3x_1 + 2x_2 = 16, \quad x_1 + 2x_2 = 12$

$\Rightarrow x = (2, 5)$: where the lines meet

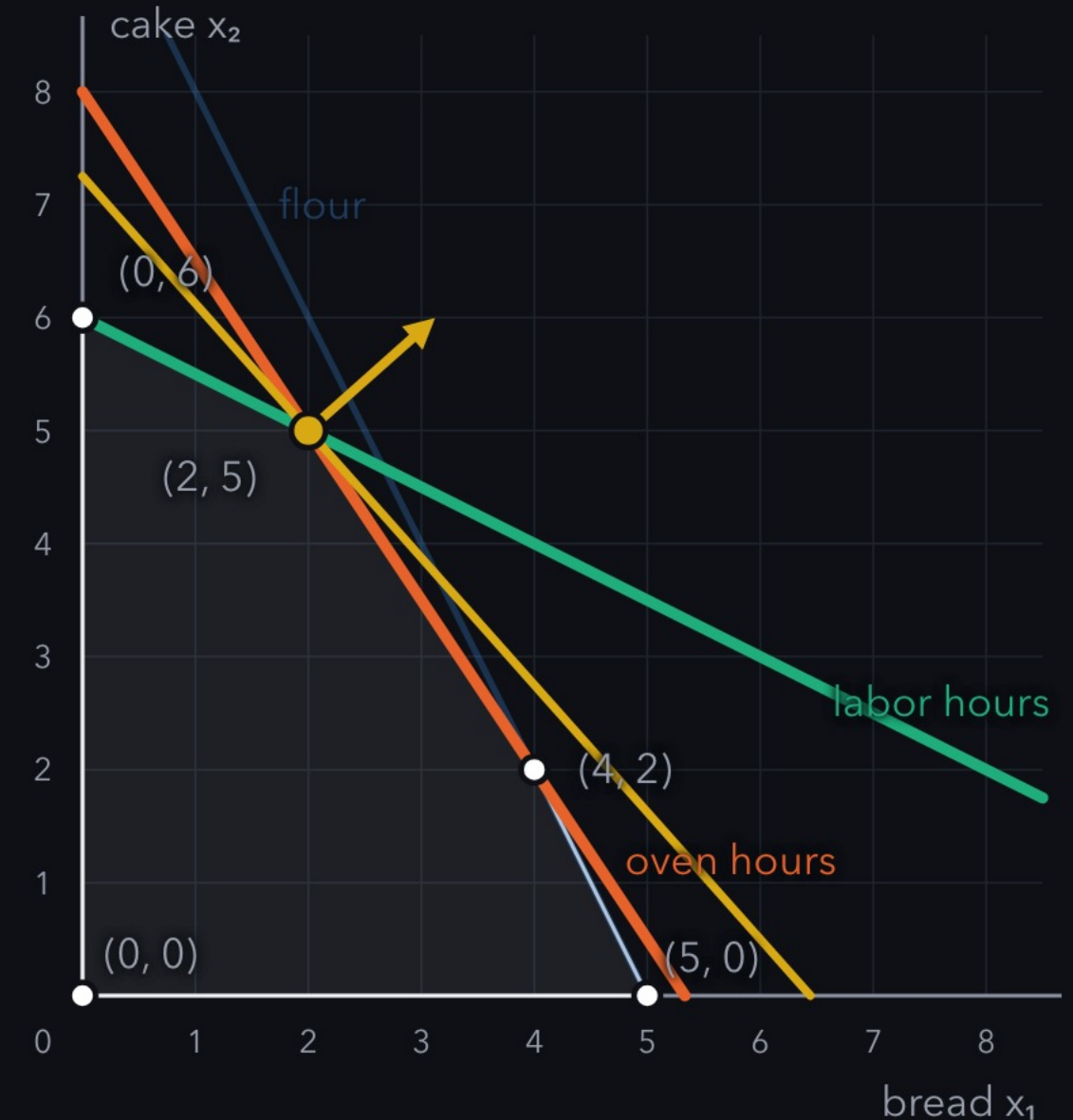
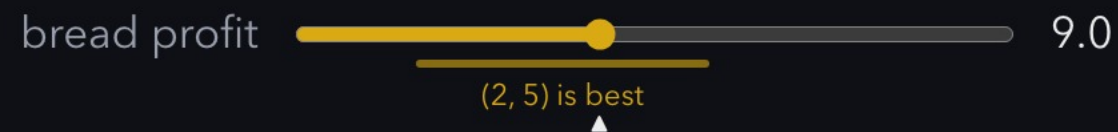
dual
by columns $3y_2 + 1y_3 = 9, \quad 2y_2 + 2y_3 = 8$

$\Rightarrow y_2 = 2.5, y_3 = 1.5$: multipliers of those lines that add up to c

In $\mathbb{R}^n$, the primal finds where n constraints meet; the dual finds multipliers of those same n constraints that add up to c . Same numbers, same value: $58 = 16 \cdot 2.5 + 12 \cdot 1.5$. When an optimal solution exists the multipliers are ≥ 0 (the profit line only touches the region), so the two views agree exactly.

$$9.0x_1 + 8x_2 - 58.0 = 2.50(\text{oven}) + 1.50(\text{labor})$$

Both weights ≥ 0 : the line only touches the region, so $(2, 5)$ is best, and the bound is its profit, 58.0.



Properties of our system at optimality

PRIMAL PERSPECTIVE

We expect that n of our primal equations are actively at an intersection (i.e., a corner point).

These constraints are tight, or at equality, meaning they should have 0 slack.

DUAL PERSPECTIVE

We expect that n of our dual multipliers are ≥ 0 .

We also expect our c vector to be exactly equal to the original, which means all constraints should have 0 slack.

Turning slack into a decision variable (and inequalities into equalities)

Slack is a big part of optimality, and we can turn it into a variable, just like our x 's

3 equations, 5 unknowns: set 2 to 0, solve for the other 3

$$2x_1 + x_2 + s_1 = 10 \text{ flour}$$

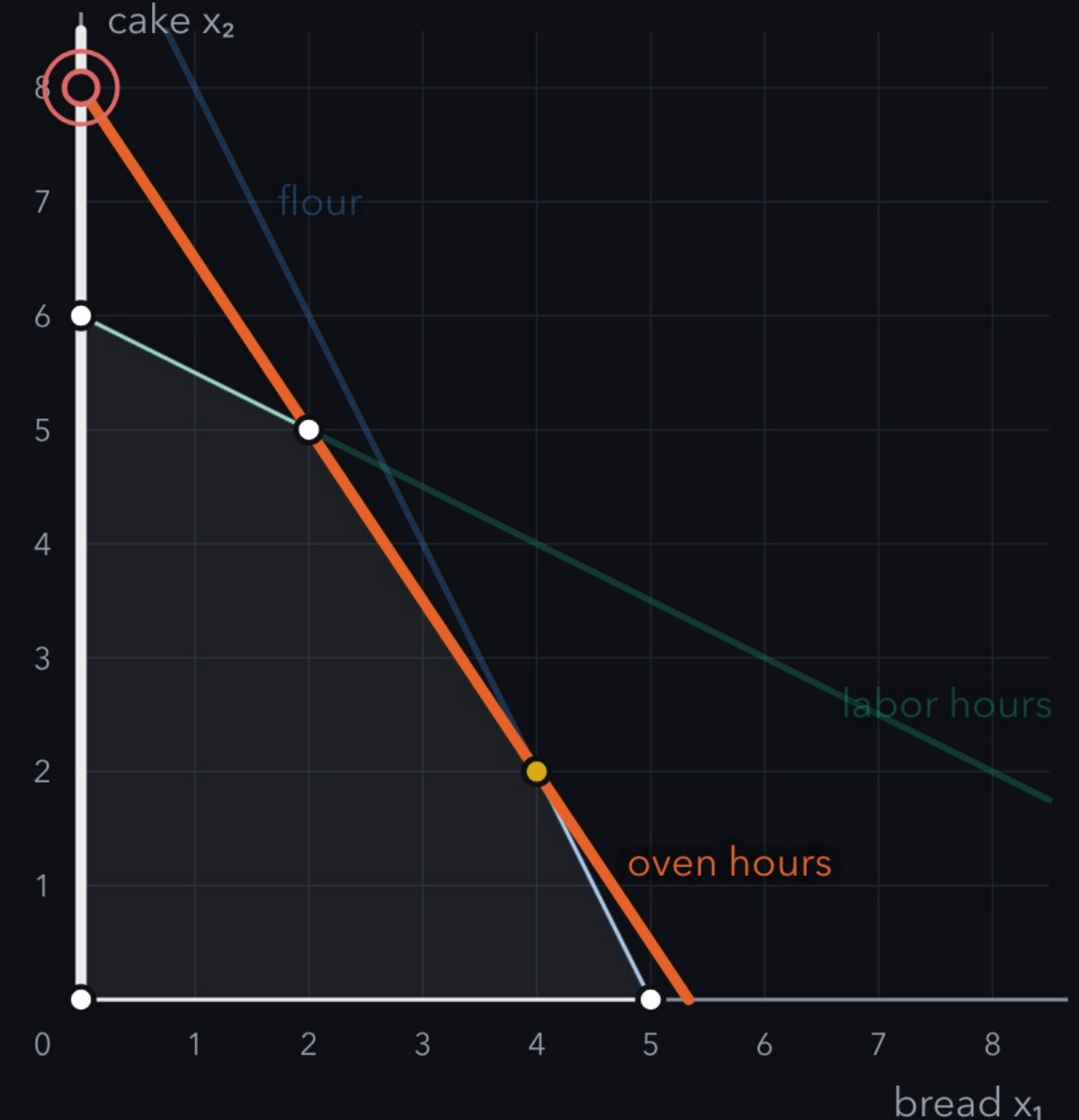
$$3x_1 + 2x_2 + s_2 = 16 \text{ oven}$$

$$x_1 + 2x_2 + s_3 = 12 \text{ labor}$$

0	8	2	0	-4	clear
≥ 0	≥ 0	≥ 0	≥ 0	≥ 0	

- Set 2 of them to 0 (here $s_1 = s_2 = 0$: flour and oven are tight), then solve the 3 equations for the other 3.
- $x_1 = s_2 = 0$ is a corner too, at $(0, 8)$, but solving gives $s_3 = -4$: labor runs over. Not a *feasible* corner.
- Try it yourself: hit **clear**, then click any 2 boxes to set them to 0; the other 3 get solved for. There are $\binom{5}{2} = 10$ corners and 5 are feasible. Find them all!

corners found: **2** of 10 · feasible (every variable ≥ 0): **1**



Dual slack: the same trick for the dual ($\geq$ into $=$)

Each dual constraint, one per product, gets a slack too

A dual slack d_j is how much the mix over-covers product j 's profit:

$$2y_1 + 3y_2 + y_3 - d_1 = 9 \text{ bread}$$

$$y_1 + 2y_2 + 2y_3 - d_2 = 8 \text{ cake}$$

$$y_1, y_2, y_3, d_1, d_2 \geq 0$$

corner $(2, 5)$, tight: oven, labor

✓ primal valid: $x = (2, 5)$, profit 58

✓ dual valid: $y = (0, 2.5, 1.5)$, bound 58

bread $d_1 = 0$ (exact)

cake $d_2 = 0$ (exact)

At each corner, y is fixed by its two tight constraints: $y^\top = c_B^\top B^{-1}$ (B : its basis, coming up with the tableau).

Inside the region the primal is valid; outside, the dual can be.

Outside bounds: 80, 64, 108, 61.33. Only $(2, 5)$ has both: $58 = 58$.



Primal and dual variables and slacks at a corner point

Every primal variable has a dual partner: x_j with its dual slack d_j , s_i with its dual variable y_i

corner **(2, 5)** · tight: oven and labor · both rows ≥ 0 : **optimal**

	x_1	x_2	s_1	s_2	s_3
PRIMAL	2	5	1	0	0
	$x_1 \cdot d_1 = 0$	$x_2 \cdot d_2 = 0$	$s_1 \cdot y_1 = 0$	$s_2 \cdot y_2 = 0$	$s_3 \cdot y_3 = 0$
DUAL	0	0	0	2.5	1.5
	d_1	d_2	y_1	y_2	y_3

- **Primal:** the corner's 2 tight constraints give 2 zeros (a slack, or an x on its axis); the other 3 are solved for: the basis.
- **Dual:** the partners of those 3 are 0; the other 2 are solved for: the multipliers of the 2 tight constraints (the dual's basis).
- In every pair at least one is 0: *complementary slackness*. The corner is optimal when both rows are ≥ 0 .



The simplex method, in one picture

The best plan is always at a corner, so walk from corner to corner

1. Start at a corner.
2. Look along the edges. Which ones improve **profits**?
3. Walk one to the next corner. Repeat.
4. Stop when no edge improves **profits**.

There was a shorter way (2 pivots vs. 3 pivots). **Which edge to take** is the *pivot rule*, and it's the question this talk builds up to.



Putting it all together: how simplex works

Relax dual feasibility. Go from corner point to feasible corner point, guided by dual variable values towards dual feasibility.

$$A = \begin{bmatrix} x_1 & x_2 & s_1 & s_2 & s_3 \\ 2 & 1 & 1 & 0 & 0 \\ 3 & 2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} x_1 \\ x_2 \\ s_1 \\ s_2 \\ s_3 \end{matrix} \begin{bmatrix} x \\ 2 \\ 5 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{matrix} c \\ 9 \\ 8 \\ 0 \\ 0 \end{matrix} \quad \begin{matrix} y \\ d_1 \\ d_2 \\ y_1 \\ y_2 \\ y_3 \end{matrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 2.5 \\ 1.5 \end{bmatrix} \quad \begin{matrix} b \\ 10 \\ 16 \\ 12 \end{matrix} \quad B = \begin{bmatrix} x_1 & x_2 & s_1 \\ 2 & 1 & 1 \\ 3 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 0 & 0.5 & -0.5 \\ 0 & -0.25 & 0.75 \\ 1 & -0.75 & 0.25 \end{bmatrix} \quad D = \begin{bmatrix} y_2 & y_3 \\ 3 & 1 \\ 2 & 2 \end{bmatrix}$$

ITERATION 4 · AT CORNER (2, 5) · OPTIMAL

basis $B = \{x_1, x_2, s_1\}$ the 3 variables we solve for; the other 2, and their 3 dual partners, are 0

primal $B x_B = b \Rightarrow x_B = B^{-1}b = (2, 5, 1)$ corner (2, 5), profit 58

dual $D (y_2, y_3)^T = (9, 8)^T \Rightarrow (y_2, y_3) = (2.5, 1.5)$ the dual's 2 equations; the partners of B are 0

check dual basics ≥ 0 ? $y_2 = 2.5, y_3 = 1.5$ all ≥ 0 : dual feasible, **optimal**

stop $c^T x = b^T y = 58$ primal and dual agree: stop

This is essentially the **revised simplex method**. Real solvers skip D and reuse B : $y^T = c_B^T B^{-1}$, because B^{-1} already holds what D needs. They also update B 's factorization at each pivot instead of rebuilding it: much cheaper on large problems.



The tableau: all the bookkeeping in one place

Bottom row: reduced costs (dual slacks) under the products, shadow prices (dual variables) under the slacks

basic	x_1	x_2	s_1	s_2	s_3	value	as an equation
s_1	2	1	1	0	0	10	$s_1 + 2x_1 + x_2 = 10$
s_2	3	2	0	1	0	16	$s_2 + 3x_1 + 2x_2 = 16$
s_3	1	2	0	0	1	12	$s_3 + x_1 + 2x_2 = 12$
z	-9	-8	0	0	0	0	$z - 9x_1 - 8x_2 = 0$

reduced costs $\mathbf{d}$
= dual slacks

shadow prices $\mathbf{y}$
= dual variables

profit

Optimal $\Leftrightarrow$ every bottom-row entry ≥ 0 : the primal is feasible and the dual is valid.
Here -9 and -8 say: not yet.



Pivoting: enter, walk the edge, leave, repeat

One step per click: entering variable, min ratio test, exiting variable, pivot

basic	x_1	x_2	s_1	s_2	s_3	value	min ratio
x_1	1	0	0	0.5	-0.5	2	
x_2	0	1	0	-0.25	0.75	5	
s_1	0	0	1	-0.75	0.25	1	
z	0	0	0	2.5	1.5	58	

reduced costs $\mathbf{d}$
= dual slacks

shadow prices $\mathbf{y}$
= dual variables

profit

Pivot. New corner (2, 5), profit 58. Every bottom-row entry ≥ 0 : **optimal**, after 3 pivots. The bottom row is the dual: shadow prices 0, 2.5, 1.5.



What if we chose the other direction?

Cake (−8) is negative too, so it can enter first: same steps, a different route

basic	x_1	x_2	s_1	s_2	s_3	value	min ratio
s_1	0	0	1	−0.75	0.25	1	
x_1	1	0	0	0.5	−0.5	2	
x_2	0	1	0	−0.25	0.75	5	
z	0	0	0	2.5	1.5	58	

reduced costs d
= dual slacks

shadow prices y
= dual variables

profit

Cake first: **2 pivots**. Dantzig's greedy choice took **3** (dashed). The biggest gain per unit isn't always the shortest route: which negative entry to pick is the **pivot rule**, and it's next.



2 pivots or 3: does it matter?

Not for our bakery. But real problems are bigger, often integer, and rarely start at a convenient corner.

Bigger problems

thousands to millions

Real LPs (airline crews, power grids, supply chains) have thousands to millions of variables and constraints. Each pivot scales with the increased problem size.

Integer problems

an LP at every node

Whole-number decisions (appointment slots, how many trucks) are solved by branch and bound, i.e., many LPs per problem, meaning a few extra pivots per LP add up fast.

Harder starts

no easy first corner

In our bakery, zero product is feasible. "At least" or "exactly" constraints (bake at least 20 batches, staff exactly 8 hours) would require simplex to run an additional phase to **find** a feasible start.

Even **one more dimension** on this toy problem changes the landscape. Next: a third product.

A new product: cookies. Worth making?

Price it with the shadow prices we already have

	bread	cake	cookies	available	price y
■ flour	2	1	1	10	0
■ oven hours	3	2	1	16	2.5
■ labor hours	1	2	1	12	1.5
profit / batch	9	8	5		

value of a batch = $1 \cdot 0 + 1 \cdot 2.5 + 1 \cdot 1.5 = 4$

reduced cost $d_3 = 4 - 5 = -1$

$$B^{-1}a_3 = B^{-1}(1, 1, 1)^T = (0, 0.5, 0.5)^T$$

Cookies' column in the tableau. B^{-1} is the highlighted slack block, so this is its row sums: per batch, cake drops 0.5, spare flour drops 0.5, bread stays put.

sub-optimal tableau for bread, cake and cookies

basic	x_1	x_2	x_3	s_1	s_2	s_3	value
x_1	1	0	0	0	0.5	-0.5	2
x_2	0	1	0.5	0	-0.25	0.75	5
s_1	0	0	0.5	1	-0.75	0.25	1
z	0	0	-1	0	2.5	1.5	58
reduced costs d = dual slacks			shadow prices y = dual variables			profit	

Cookies add a dimension

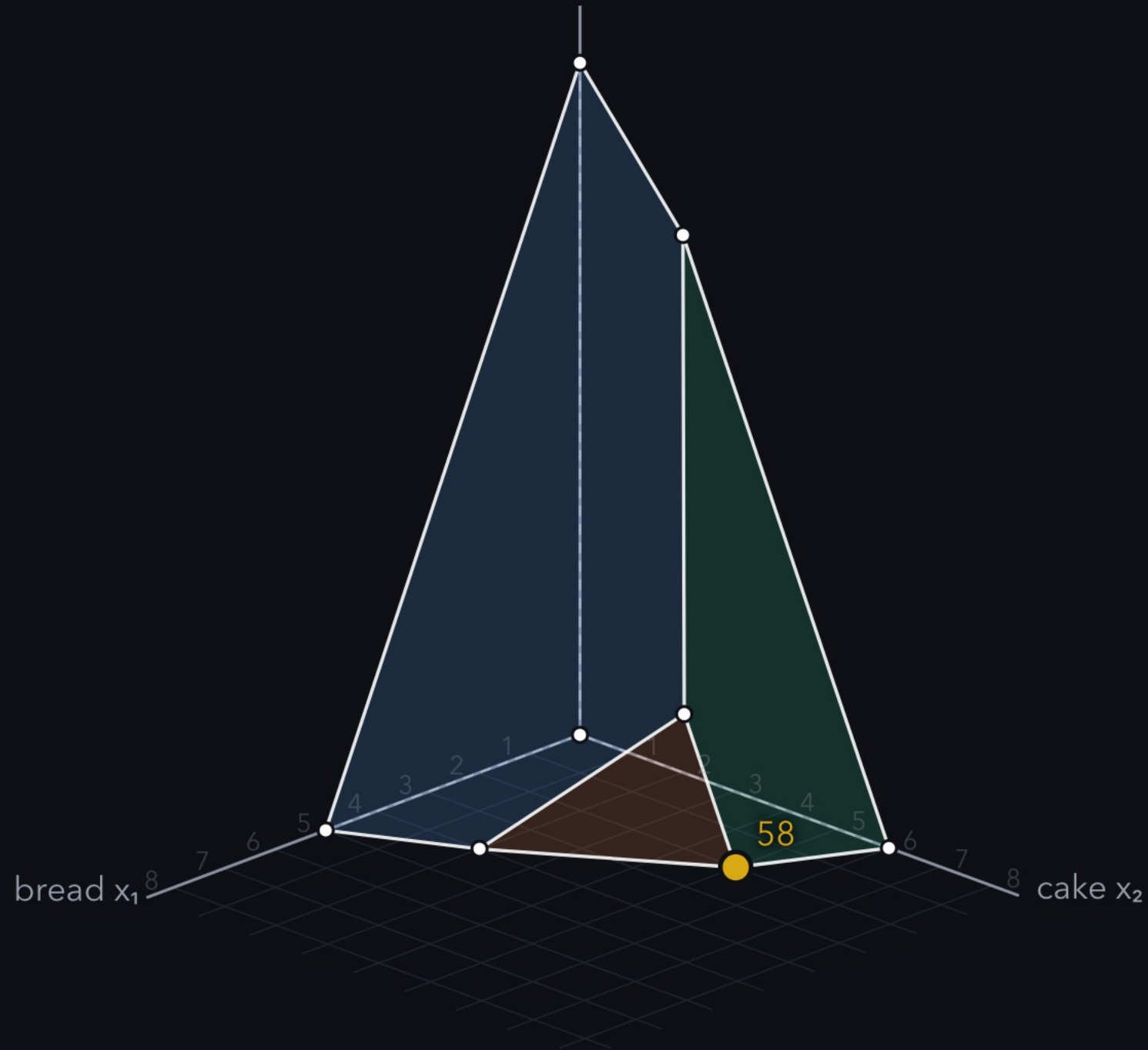
Our pentagon, seen from above.

Tilt the camera: it is the **floor** of a 3D picture, where $x_3 = 0$ (no cookies).

Each constraint becomes a **plane**, and the plans form a polyhedron with 8 corners.

The 2D picture was the floor all along.

Drag the picture to rotate it.



Slide the profit plane

In 3D, all plans with the same profit lie on a plane

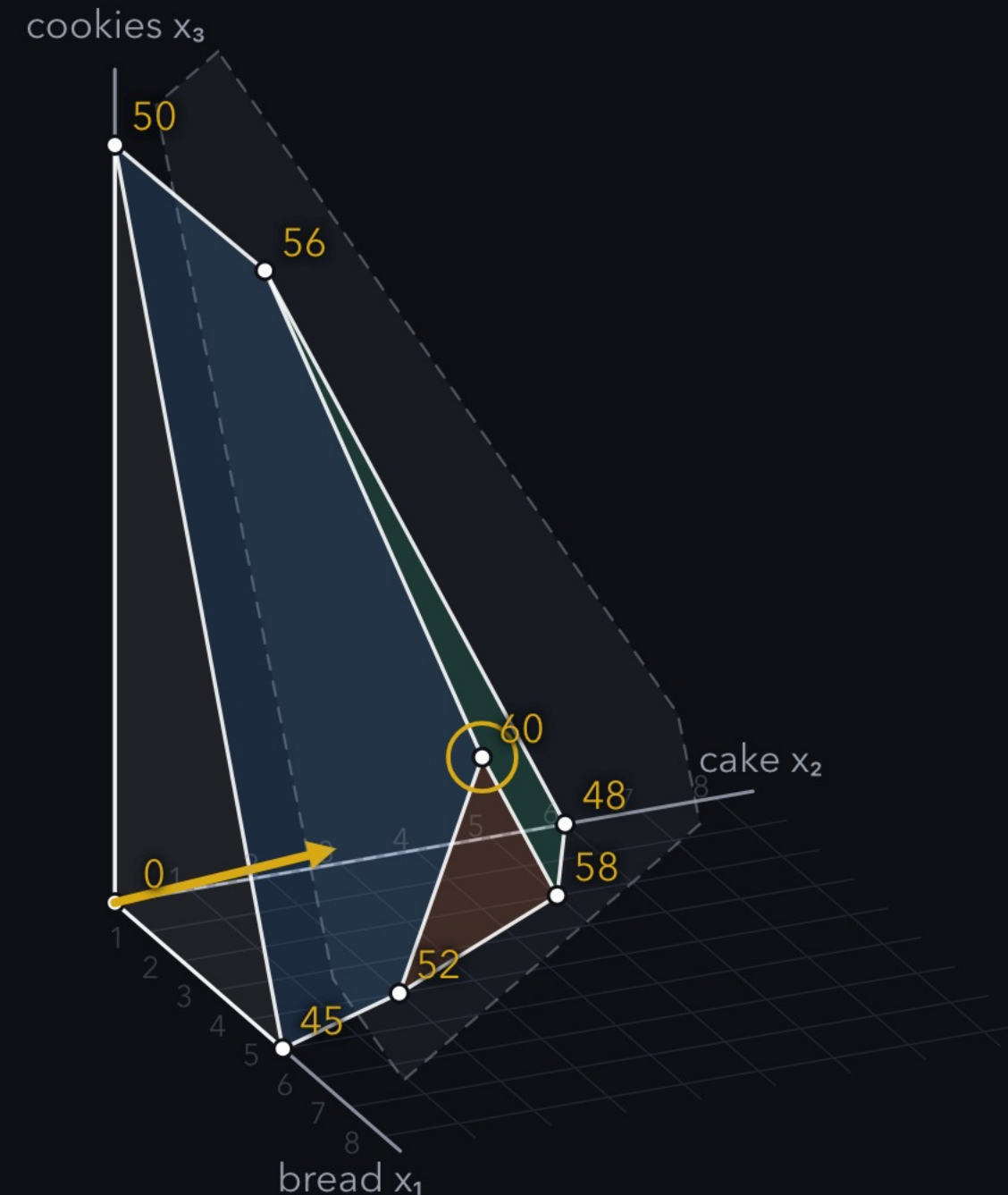
$$9x_1 + 8x_2 + 5x_3 = z$$

$z = 66$: no plan

Push the plane along the **arrow**. The gold slice is every plan that earns exactly z .

The slice shrinks to a single **corner**: $(2, 4, 2)$, profit **60**.

Any further and no plan is left.

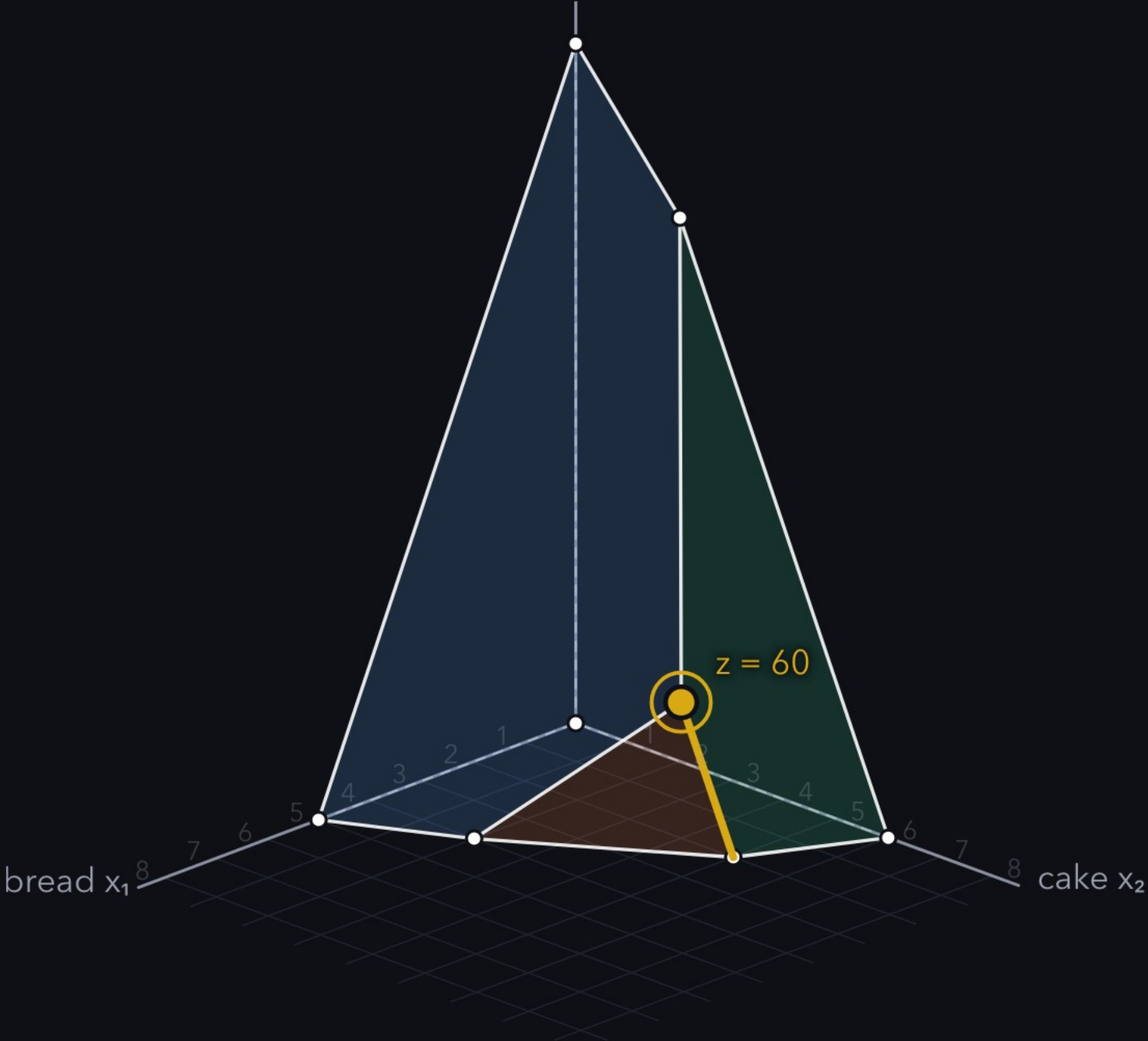


One pivot up

Cookies enter; walk up the edge until a resource runs out

basic	x_1	x_2	x_3	s_1	s_2	s_3	value
x_1	1	0	0	0	0.5	-0.5	2
x_2	0	1	0	-1	0.5	0.5	4
x_3	0	0	1	2	-1.5	0.5	2
z	0	0	0	2	1	2	60
reduced costs $\mathbf{d}$ = dual slacks				shadow prices $\mathbf{y}$ = dual variables			profit

Every entry ≥ 0 : **optimal**. All three resources are used up, so all three have a price: 2, 1, 2.



The 3D dual: three planes meet at the corner

$$2 \times (2x_1 + x_2 + x_3 \leq 10)$$

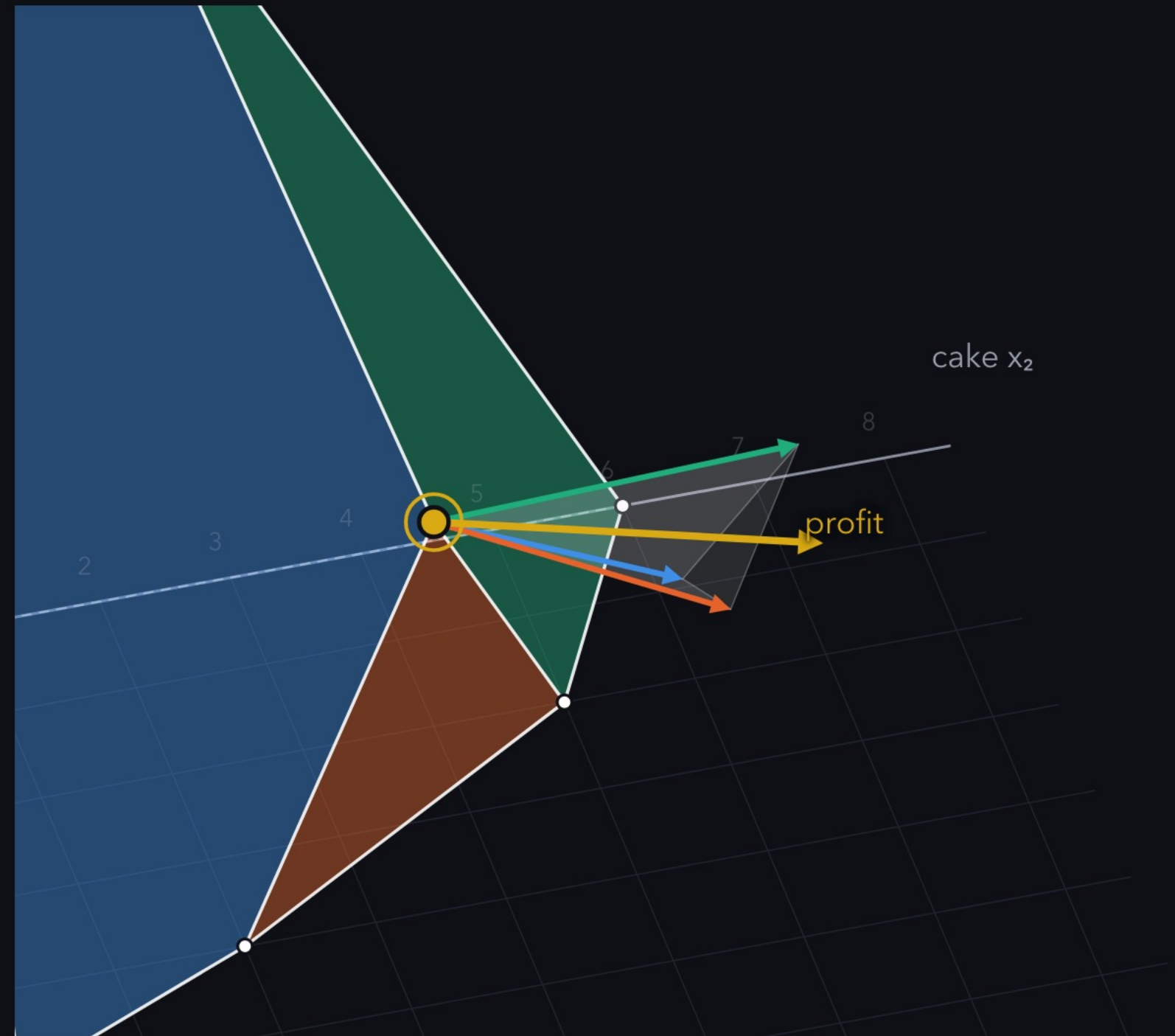
$$1 \times (3x_1 + 2x_2 + x_3 \leq 16)$$

$$2 \times (x_1 + 2x_2 + x_3 \leq 12)$$

$$= 9x_1 + 8x_2 + 5x_3 \leq 60$$

That is exactly the profit $9x_1 + 8x_2 + 5x_3$, so no plan beats 60, and $(2, 4, 2)$ earns 60.

Three tight planes, three unknowns $\Rightarrow$ one corner. The **profit arrow** lies inside the cone of the planes' outward directions $\Leftrightarrow$ every multiplier $\geq 0 \Leftrightarrow$ optimal.



One corner, four rules

At the origin, bread, cake and cookies all have negative reduced costs. Which one enters?

	bread x_1	cake x_2	cookies x_3
reduced cost d	-9	-8	-5
Dantzig : most negative d	-9	-8	-5
step until a resource runs out t^*	5	6	10
Greatest increment : biggest gain $ d t^*$	45	48	50
edge length ² $1 + \ B^{-1}a_j\ ^2$	15	10	4
Steepest edge : d^2 / length^2	5.40	6.40	6.25
Bland : lowest index	first		

The edge length counts all six variables, leftovers included: a batch of bread uses up a lot of resources.

Three different answers at the same corner. Each rule is a different guess at which edge leads to the top fastest.

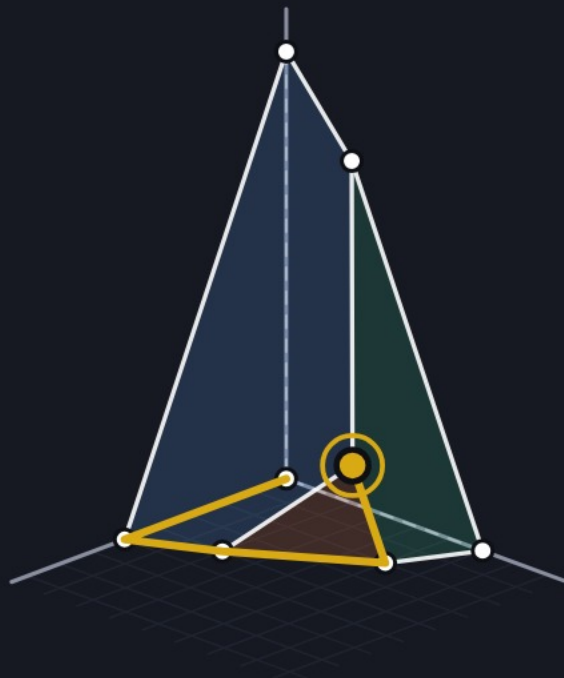


The race: same polyhedron, four routes

Every rule reaches the same optimum (2, 4, 2), by a different path

Dantzig

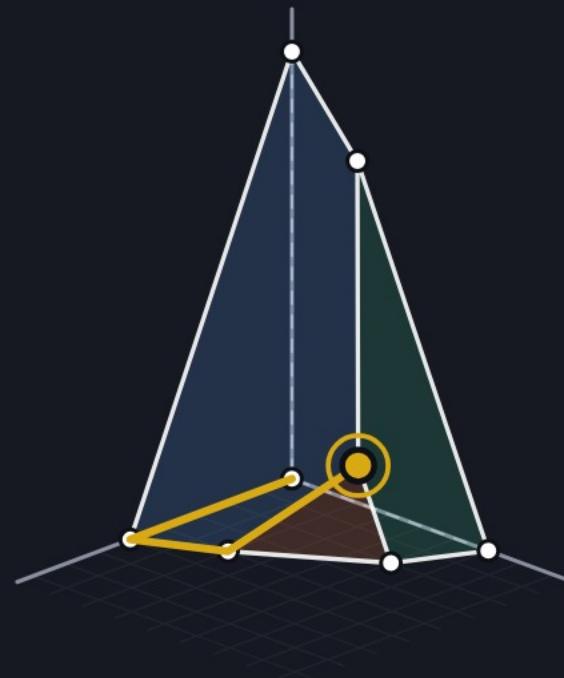
most negative d



4 pivots ✓

Bland

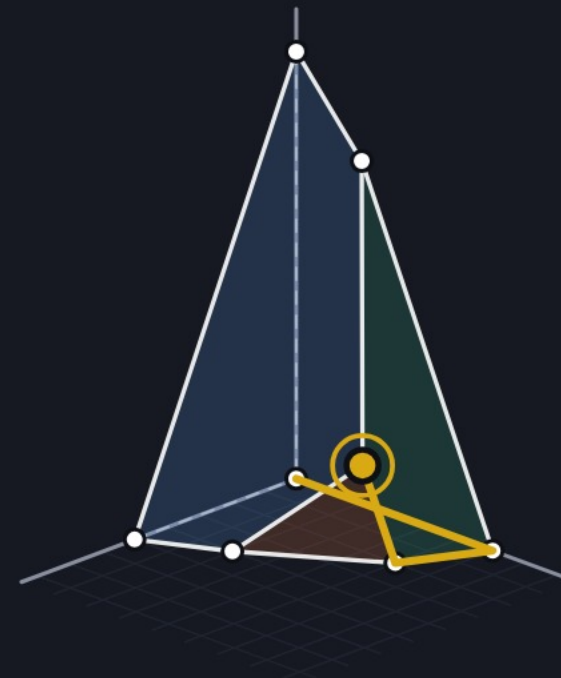
lowest index



3 pivots ✓

Steepest edge

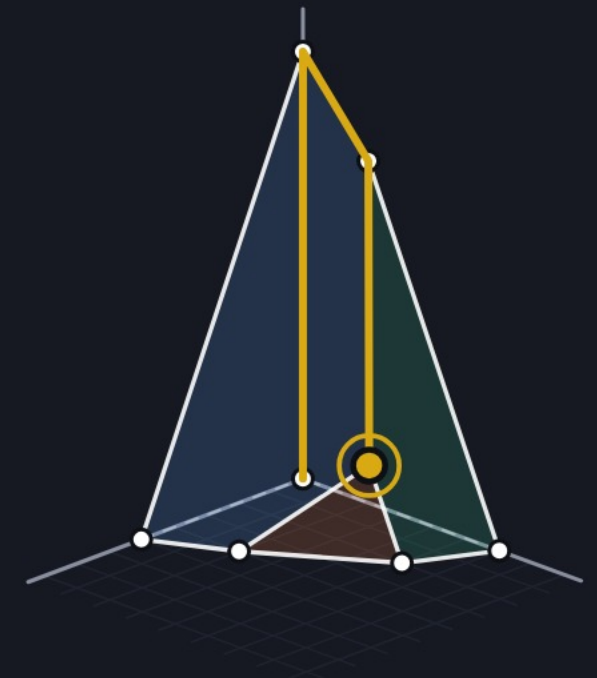
best gain per unit length



3 pivots ✓

Greatest increment

biggest gain this step



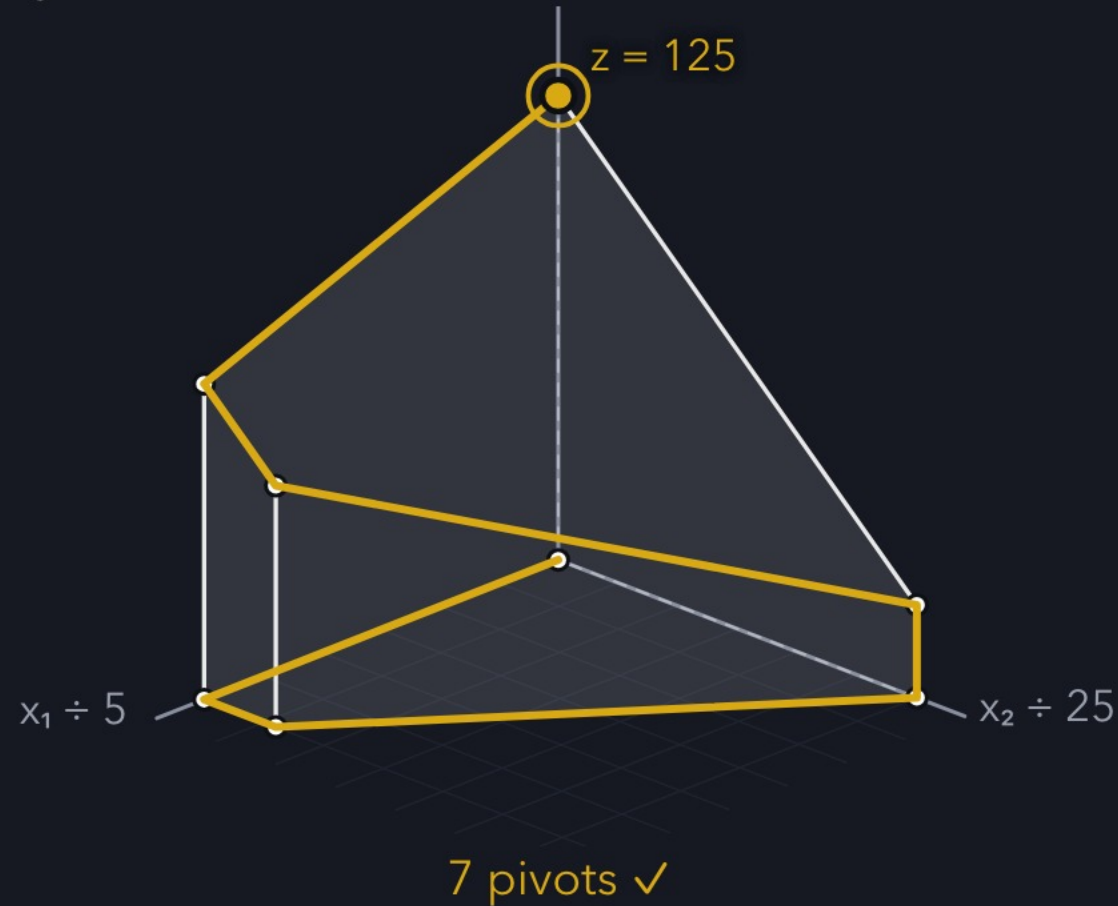
3 pivots ✓

Dantzig solves the bread-and-cake problem on the floor first, then climbs: **4 pivots**. The others need **3**, the fewest possible here (there are four different shortest routes). On a problem this small the gap is one pivot. It can be much worse...

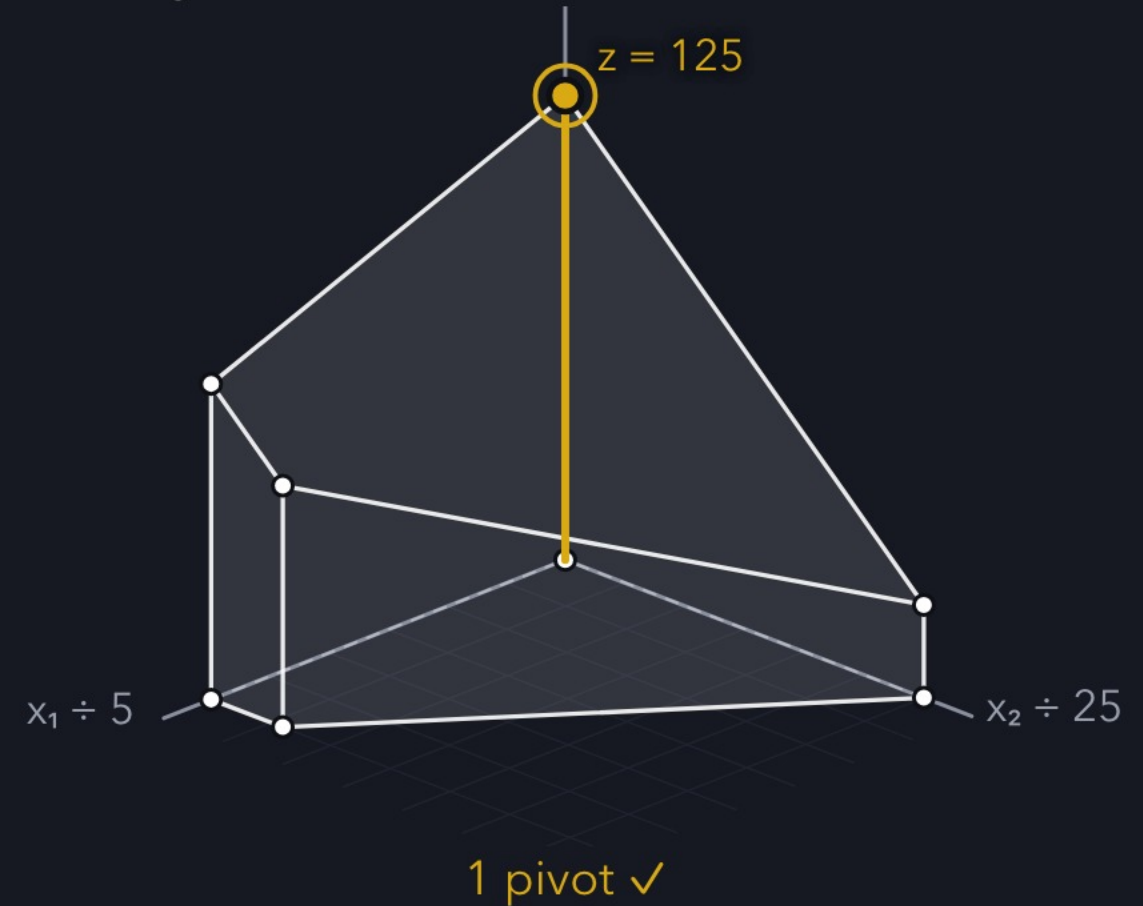
When the rule really matters: the Klee-Minty cube

$\max 4x_1 + 2x_2 + x_3$ s.t. $x_1 \leq 5$, $4x_1 + x_2 \leq 25$, $8x_1 + 4x_2 + x_3 \leq 125$ (axes rescaled to look like a cube)

Dantzig



Steepest edge



Dantzig visits **all 8 corners** (7 pivots); steepest edge goes straight to the top (1 pivot). The same construction in n dimensions makes Dantzig take $2^n - 1$ pivots: about 10^{15} in 50 dimensions. (Bland: 5 here.)

Two more things about rules

Devex

Steepest edge needs the length of every candidate edge at every pivot, which is expensive on large problems. Devex keeps cheap running *estimates* of those lengths instead.

Real solvers use Devex or steepest edge. On the bakery, Devex makes the same choices as Bland.

Bland's rule and cycling

At a *degenerate* corner (more constraints tight than needed), a pivot can change the basis without moving. A careless rule can then cycle through the same bases forever.

Bland's rule provably never cycles. It is slow in practice, so it is mostly a safety net; solvers also break ties with small perturbations.

For our research, we can depict rules as a pivot timeline

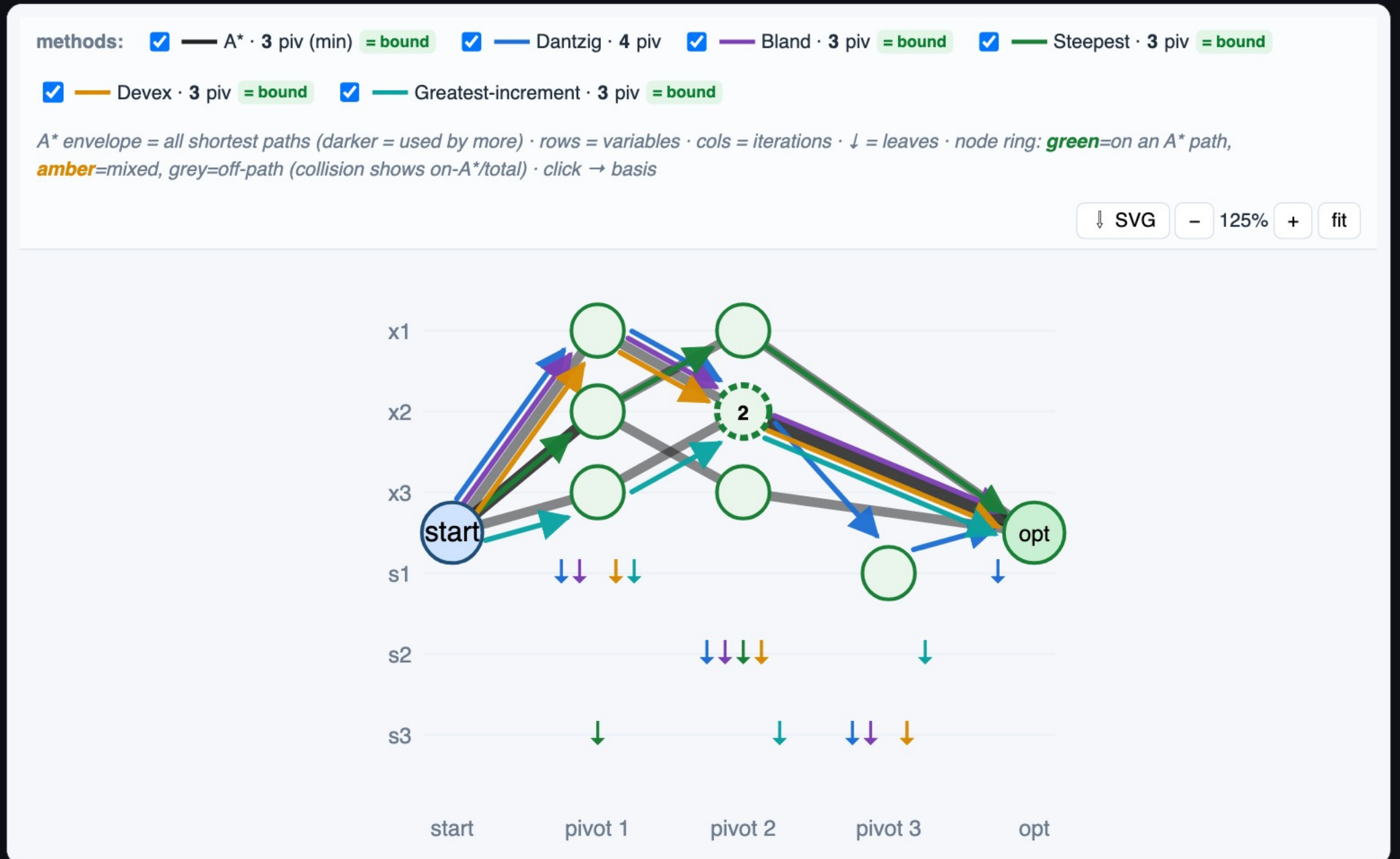
All previous routes without the polyhedron (meaning it can generalize for much larger LPs)

Columns are pivots: start → ... → optimum.

Rows are variables. A node is the variable that **enters** at that pivot; ↓ marks the one that **leaves**.

Each coloured thread is one rule. Dantzig's needs an extra column.

A* (black) searches every option to find a provably shortest route. Far too slow to run inside a solver, but it shows how short a route *can* be.



Klee-Minty, as a timeline

Dantzig's thread snakes through 7 columns; steepest edge and A* jump straight to the optimum



Our question: can a policy *learn* to pick pivots?

Pivots needed on an LP with 40 products and 40 resources



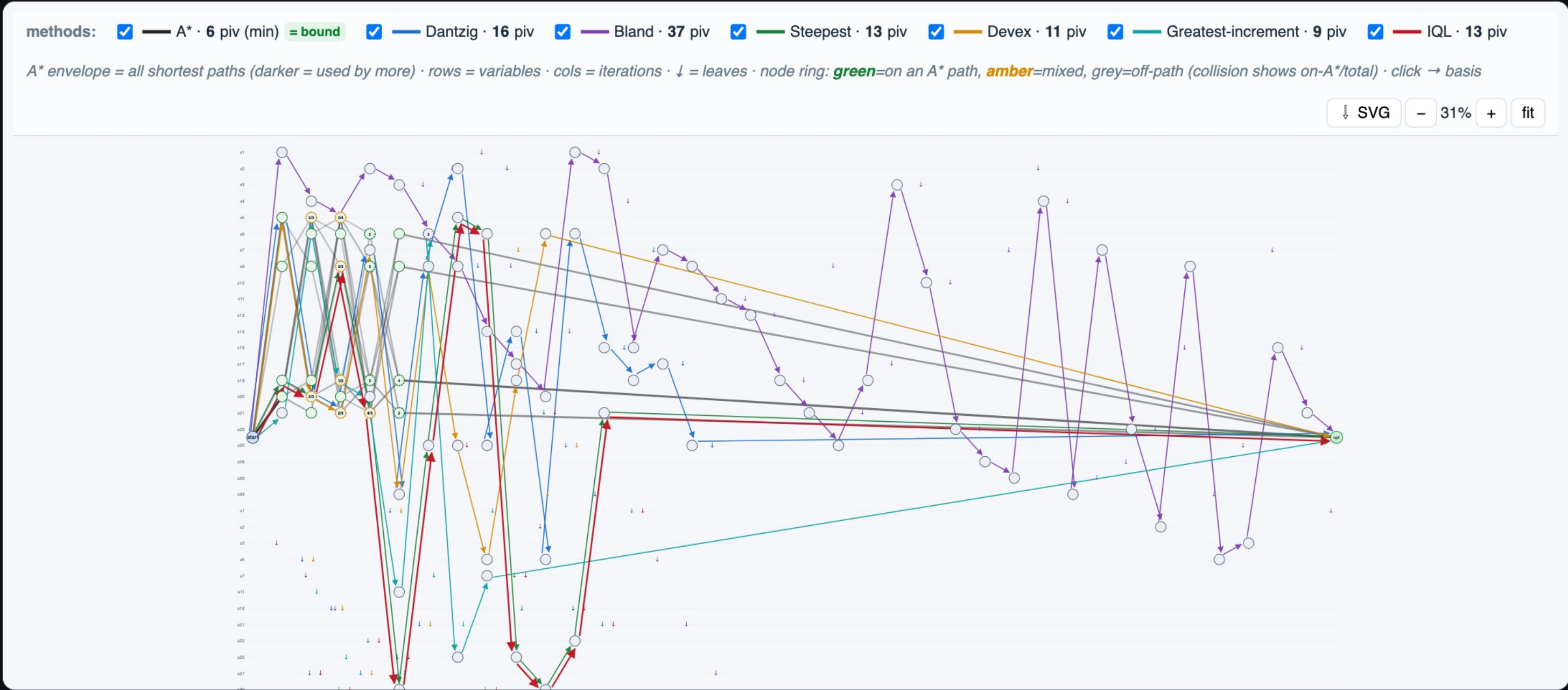
A short route **exists**: 6 pivots. A* proves it, but only by searching far more than a solver can afford.

Every rule is a **fixed guess** about which edge to take. The best one here needs 9; Bland needs 37.

SimplexRL: learn the guess. A neural policy looks at the current tableau, picks the entering variable, and is rewarded for reaching the optimum in fewer pivots.

40 products, 40 resources: no picture, same timeline

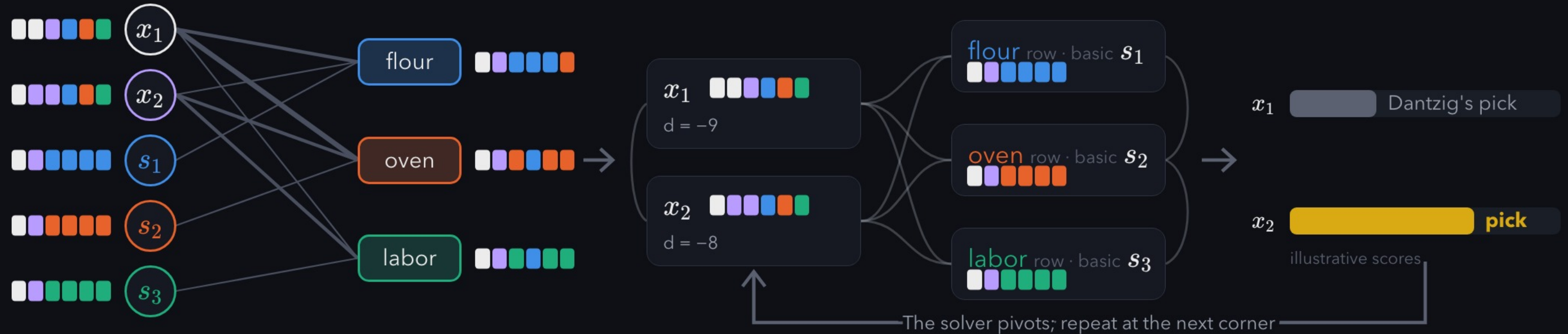
An 80-dimensional polyhedron we can't draw, but every route is still visible



Inside SimplexRL: a network that learns to pivot *in progress*

Our current model, walked through on the bakery

Felix Parker, Emmett Springer, Danielle Ripsman, Kimia Ghobadi



1 · Read the LP once per problem

Variables and constraints become nodes; each nonzero of A is an edge (thicker = bigger coefficient).

A **graph neural network** passes messages along the edges; each coloured strip is a node's vector (embedding).

2 · Compare the candidates every pivot

Each candidate (negative reduced cost) adds the solver's numbers; a **transformer** compares candidates with each other and with the current basis.

3 · Pick

The solver pivots on the top score. Dantzig takes bread: 3 pivots. Cake first: 2.

How it learns

Imitate a classic rule, or A^*

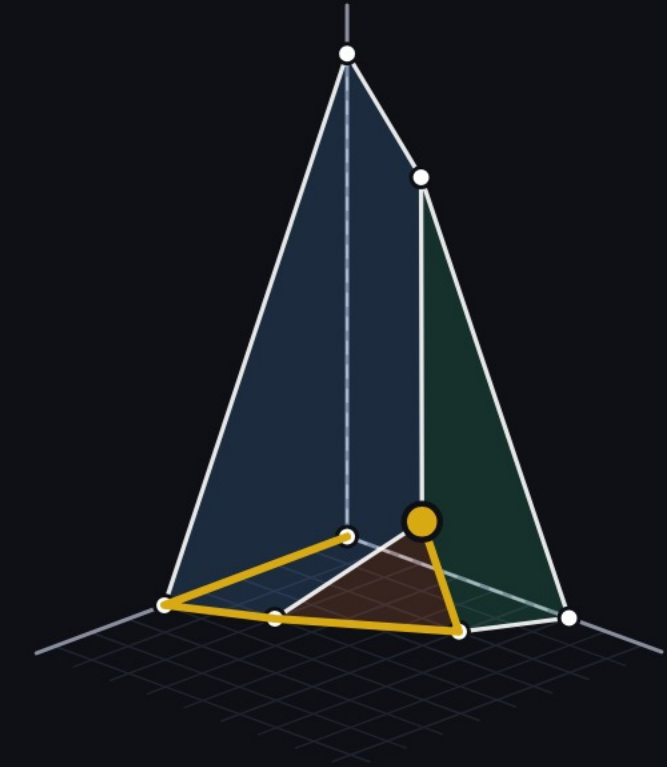
Reinforcement learning: -1 per pivot

Search as a teacher (like AlphaZero): in training, try pivots ahead in the solver and teach the network the best one

The early learned offline policy (IQL) already ties steepest edge on the 40×40 LP.

Takeaways

- 1 The best plan sits at a **corner** (or on a tied edge or face, whose corners are all best too).
- 2 **Shadow prices are dual variables**: adding up the tight constraints with them proves a plan optimal.
- 3 **Reduced costs are dual slacks**: a negative one says the dual doesn't work yet, and which way to move.
- 4 A **pivot rule** is actually just a policy that we can decide. The 'best' one can vary problem to problem, potentially creating or removing exponential amounts of work.



Thank you! Questions?
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optimal-lab.com/talks/walking-the-corners